Practice workbook · What the operating system does, and system vs application software

What the operating system does (process, memory, file, peripheral and user-interface management, multitasking, file storage on disc) and the difference between system and application software.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-12

1. Question 5 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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1. Question 5 · part e · 2 marks

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Worked-example retry · Attempt part e only.

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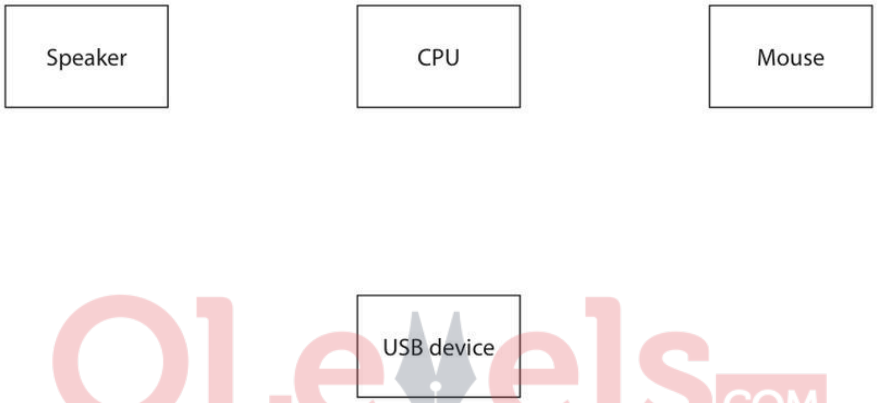
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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1. Question 5 · part e · 2 marks

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Worked-example retry · Attempt part e only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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1. Question 5 · part e · 2 marks

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Worked-example retry · Attempt part e only.

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(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

(2)

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

(3)

A	B	C	A AND C	(A AND C) OR B
0	0	0	0	0
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1	1	1

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P 7 2 9 3 7 A 0 1 7 2 4

17

Turn over ▶

1. Question 5 · part e · 2 marks

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Worked-example retry · Attempt part e only.

(ii) A warehouse has an automated alarm system.

When the alarm system is activated it will sound if:

- a movement sensor (M) is activated
- a pressure pad (P) is activated
- a key code (C) to deactivate the alarm system has not been entered.

Construct a logic statement, using AND, OR and NOT with the letters M, P and C, to show the conditions that will sound the alarm.

(3)

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1. Question 5 · part e · 2 marks

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Worked-example retry · Attempt part e only.

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(e) A computer with a single CPU runs several processes at the same time.

This computer is multitasking.

Describe how the operating system enables processes to share a single CPU.

(2)

(Total for Question 5 = 18 marks)

Working and answer

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2. Question 1 · part f · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part f only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

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2 .....

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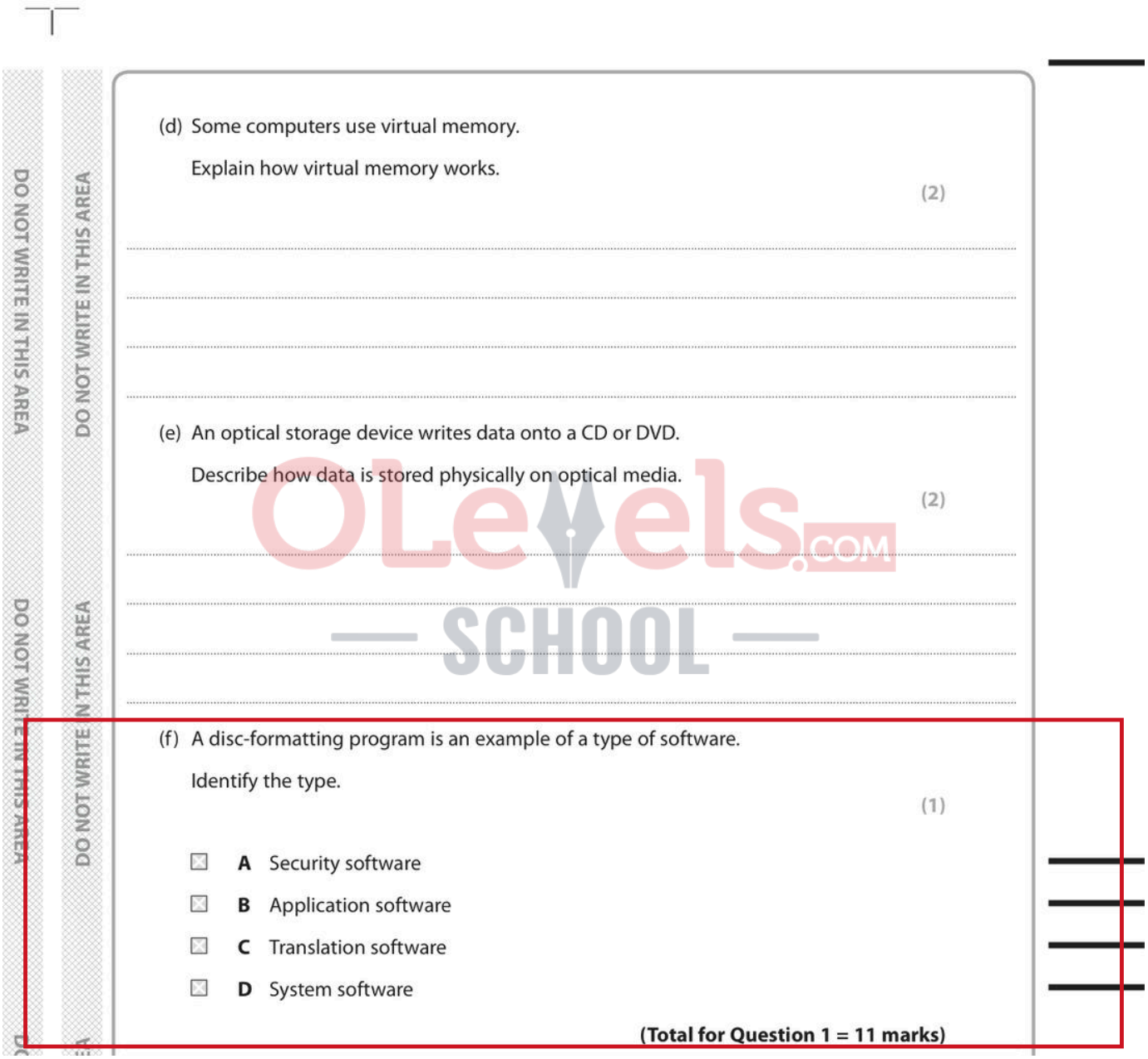
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2. Question 1 · part f · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part f only.



(d) Some computers use virtual memory.

Explain how virtual memory works.

(2)

(e) An optical storage device writes data onto a CD or DVD.

Describe how data is stored physically on optical media.

(2)

(f) A disc-formatting program is an example of a type of software.

Identify the type.

(1)

- ☒ **A** Security software
- ☒ **B** Application software
- ☒ **C** Translation software
- ☒ **D** System software

(Total for Question 1 = 11 marks)

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2. Question 1 · part f · 1 mark · Working and answer

Working and answer

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3. Question 3 · part d · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

3 The central processing unit (CPU) and other hardware components work together with the operating system to run programs and store data.

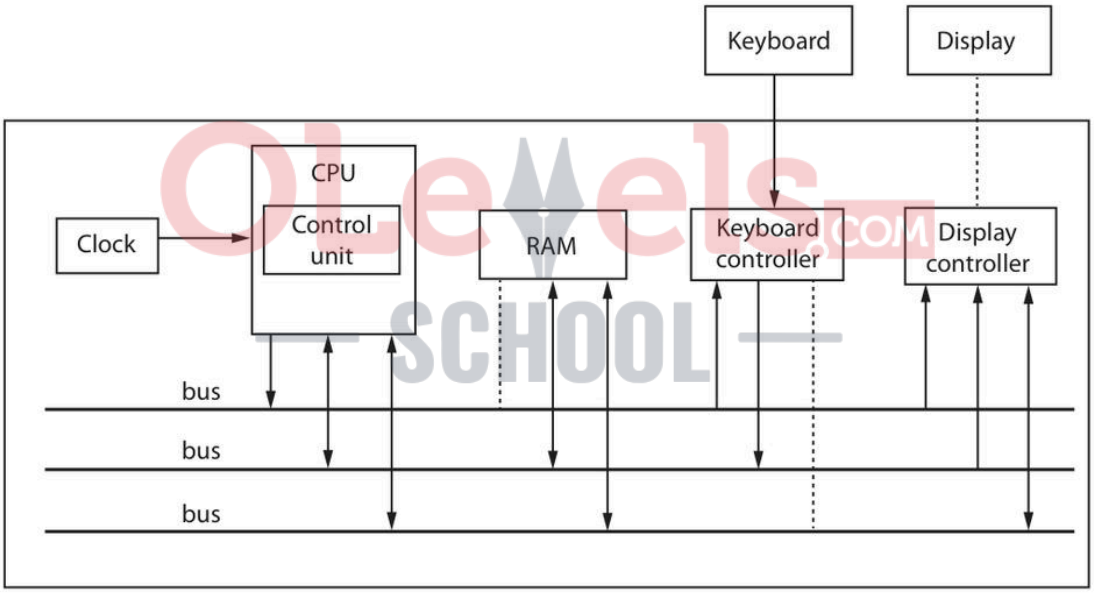
(a) Here is a partially completed diagram showing the hardware components used in the fetch-decode-execute cycle.

The function of the I/O controller is to manage communication between the CPU and input/output devices.

Complete the diagram by:

- labelling the buses
- adding directional arrows to the dotted lines to show the flow of communication.

(6)



(b) The CPU includes an area of cache memory.

Increasing the size of the cache will improve the performance of the CPU.

Explain how increasing the size of the cache improves the performance.

(2)

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3. Question 3 · part d · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

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(c) Some computers have more than one CPU. The CPUs work in parallel to execute instructions.

Complete the table to show how two CPUs working in parallel could execute the expression:

(5 * 2) + (4 * 3) – (24 / 4)

(4)

CPU 1	CPU 2
5 * 2 = 10	

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(d) One function of an operating system is to manage files on secondary storage.

Describe how an operating system manages the storage of a file on random access secondary storage.

(4)

(Total for Question 3 = 16 marks)

3. Question 3 · part d · 4 marks · Working and answer

Working and answer

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4. Question 3 · part c · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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P 7 2 9 3 7 A 0 8 2 4

4. Question 3 · part c · 2 marks

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Further practice · Attempt part c only.

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

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(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

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P 7 2 9 3 7 A 0 9 2 4

9

Turn over ▶

4. Question 3 · part c · 2 marks · Working and answer

Working and answer

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5. Question 2 · part a.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a.i only.

2 Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

(1)

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(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

(3)

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5. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

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6. Question 2 · part a.ii · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part a.ii only.

2 Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

(1)

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

(3)

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6. Question 2 · part a.ii · 3 marks

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Further practice · Attempt part a.ii only.

(b) Computers use different types of storage.

(i) Explain **one** reason why computers need both RAM and ROM.

(2)

(ii) Describe how data is read from a magnetic hard drive.

(3)

(c) Computers perform the fetch-decode-execute cycle.

Complete the table by writing the numbers 1 to 4 in the sequence number column to put each stage in the correct order.

Stage	Sequence number
The instruction is placed onto the data bus	
The Arithmetic and Logic Unit (ALU) performs a calculation	
The address of the next instruction is placed onto the address bus	
The instruction is decoded by the control unit	

(4)

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P 7 5 7 3 6 A 0 7 2 8

6. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

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